

1 Primitive response-scale growth control

Source: PrimeTensor/Analysis/Growth.lean

What this file establishes

PrimeTensor/Analysis/Chain.lean proved the chain rule under a hypothesis, ChainAdmissibleAt, that quantifies over *arbitrary error germs*: it says that anything negligible against the inner response is negligible against the perturbation. That is a strong and rather indirect thing to have to assume, and the module header names the defect precisely — it is a second-order transport assumption where a first-order geometric one ought to do.

This file supplies the geometric condition. ResponseScaleControlledAt speaks only about the perturbation and its response, quantifying over no error germs at all, and it implies admissibility. The chain rule is then restated with it in place of the transport hypothesis, and the two base cases — the identity and the constants — are verified.

The interesting part is how a growth bound gets written down at all, since Depth has no subtraction. The header states the technique: scale comparison is expressed entirely through AtOrAfter. Design note 1.2 works out what that buys.

1.1 The condition

Definition 1.1 (ResponseScaleControlledAt). f is *response-scale controlled at x* when, for every perturbation h approaching the pivot and every target gain t , there are a response gain r and an anchor such that, beyond the anchor and for every level ℓ the perturbation achieves, the response achieves *some* level ρ with

$$\text{ScaleNear}(\rho, \text{response}(f, x, h)_n, 1) \quad \text{and} \quad \text{AtOrAfter}(\text{advance}(\ell, t), \text{advance}(\rho, r)).$$

```
def ResponseScaleControlledAt
  (f : MulReal → MulReal) (x : MulReal) : Prop :=
  ∀ h : MulReal.Seq,
  ApproachesPivot h →
  ∀ targetGain : Depth,
  ∃ responseGain : Depth,
  ∃ anchor : Depth,
  ∀ n : Depth,
  Depth.AtOrAfter anchor n →
  ∀ level : Depth,
  MulReal.ScaleNear level (h n) 1 →
  ∃ responseLevel : Depth,
  MulReal.ScaleNear responseLevel
    (response f x h n) 1 ∧
  Depth.AtOrAfter
    (Depth.advance level targetGain)
    (Depth.advance responseLevel responseGain)
```

Design note 1.2 (A growth bound without subtraction). Read the concluding comparison numerically. AtOrAfter(a, b) is $|a| \leq |b|$ and $|\text{advance}(\ell, g)| = |\ell| + |g|$, so the condition is

$$|\ell| + |t| \leq |\rho| + |r|, \quad \text{that is} \quad |\rho| \geq |\ell| + |t| - |r|.$$

The response's level lags the perturbation's by at most the fixed amount $|r| - |t|$. In the logarithmic chart of `PrimeTensor/Analysis/Derivative.lean`, where level ℓ is the tolerance $2^{-(\ell-1)}$, that reads

$$|\log \text{response}_n| \leq 2^{|r|-|t|} \cdot |\log h_n| \quad \text{up to the dyadic rounding,}$$

which is the classical comparison $\Delta f = O(h)$ with $2^{|r|-|t|}$ as the constant.

The inequality $|\rho| \geq |\ell| + |t| - |r|$ cannot be stated directly: Depth carries no subtraction, and none could be added, since the depths start at one and a difference need not be positive. The definition therefore never forms it. It advances *both* sides — the base level by t , the response level by r — and compares the two results in the tail order of `PrimeTensor/Completion.lean`, which is available. Moving the deficit to the other side of the comparison turns a subtraction into an addition, and addition of depths is what advance already is.

This is worth setting against `PrimeTensor/Analysis/Chain.lean`, which introduced admissibility as a hypothesis on the ground that the required big- O comparison had no expression in the vocabulary. That was true of the vocabulary as used there — a single level advanced by a gain, which can only be made finer — but not of the vocabulary as a whole. Advancing both levels and comparing recovers exactly the missing relation, which is what this file is for.

Remark 1.3 (Where each quantifier sits). The response gain r and the anchor are chosen after the perturbation and the target gain but before the stage n and the base level ℓ , so a single r must serve at every stage and every level — the constant in the bound is uniform. The response level ρ , by contrast, is chosen after ℓ , as it must be, since how fine the response is depends on how fine the perturbation is. And the condition is stated per perturbation: a function may be response-scale controlled with different constants along different h .

1.2 It implies admissibility

Theorem 1.4 (`chainAdmissible_of_responseScaleControlled`). *For every f and x ,*

$$\text{ResponseScaleControlledAt}(f, x) \longrightarrow \text{ChainAdmissibleAt}(f, x).$$

Proof. Let h approach the pivot, let error be negligible relative to the response, and let a target gain t be requested. Take the response gain r and the control anchor supplied by the hypothesis at t , take the anchor supplied by negligibility *at the gain r* , and join the two.

Beyond the join, suppose the base achieves level ℓ at stage n . The control produces a response level ρ at which the response is near the pivot, together with the comparison `AtOrAfter(advance( $\ell, t$ ), advance( $\rho, r$ ))`. Feeding ρ to the negligibility hypothesis — legitimate, since the response is near the pivot at ρ — puts error_n near the pivot at `advance( $\rho, r$ )`. The comparison says that level is at or after `advance( $\ell, t$ )`, so weakening once by `scaleNear_weaken` of `PrimeTensor/Analysis/Control.lean` delivers the required `advance( $\ell, t$ )`. $\square$

Remark 1.5 (Monotonicity, spent a second time). The last step is the whole reason the comparison is stated in the tail order rather than as an equation. Having proved the error at some level, the proof needs it at a possibly coarser one, and that is precisely the nesting of levels established in `PrimeTensor/Analysis/Control.lean`. The pattern is now familiar: a bound is obtained at whatever level the machinery produces, and monotonicity converts it to the level that was asked for.

1.3 The two base cases

Lemma 1.6 (`responseScaleControlledAt_identity` and `responseScaleControlledAt_constant`). *The identity map, and every constant map, are response-scale controlled at every point.*

Proof. In both cases take the response gain equal to the target gain, the anchor one, and the response level equal to the base level ℓ ; the comparison is then between $\text{advance}(\ell, t)$ and itself, discharged by `here`.

What differs is the first component. For the identity the response at stage n is $\text{ratio}(xh_n, x) = h_n$, so it is near the pivot at ℓ exactly because the perturbation is — the hypothesis is handed back unchanged. For a constant the response is $\text{ratio}(c, c) = 1$, near the pivot at every level by reflexivity, and the hypothesis is discarded. $\square$

The docstrings distinguish the two as *exact* and *maximal* control, and the distinction is real even though the witnesses coincide: the identity attains the bound with constant 1 and could not do better, whereas the constant map would satisfy the condition for any response gain whatever, its response being the pivot on the nose.

1.4 The chain rule, restated

Theorem 1.7 (`hasMulDerivativeAt_comp_of_responseScaleControlled`). *Suppose f has derivative D_f at x with D_f scale controlled, f is response-scale controlled at x , and g has derivative D_g at $f(x)$ with D_g scale controlled. Then $g \circ f$ has derivative $\text{comp}(D_g, D_f)$ at x .*

Proof. Convert the third hypothesis by Theorem 1.4 and apply the chain rule of `PrimeTensor/Analysis/Chain.lean`. $\square$

Remark 1.8 (What has changed and what has not). All five hypotheses are now first-order conditions on f , g and their tangent maps: two are boundedness of a tangent map, two are differentiability, and the fifth is a growth bound on a response. None quantifies over error germs. That is the whole improvement, and it is a real one — an assumption about all possible errors has become an assumption about one specific germ.

The underlying proof is unchanged; `ChainAdmissibleAt` remains the notion the chain rule actually consumes, and this theorem is a wrapper. No converse is proved, so the two conditions are not known to be equivalent, and nothing rules out an f that is admissible without being response-scale controlled.

Remark 1.9 (Still no nontrivial instance). Response-scale control is verified for the identity and for constants and for nothing else. It is not shown closed under products, inverses, powers or composition, and neither is `ChainAdmissibleAt`. So Theorem 1.7 can still be discharged only where `PrimeTensor/Analysis/Chain.lean` could — with f the identity or, now, a constant — and in both of those the composite’s derivative was already known from `PrimeTensor/Analysis/Examples.lean`. The closure lemmas that would make the rule usable are not in this file.

Summary of the correspondence

Lean declaration	Kind	Here
<code>ResponseScaleControlledAt</code>	definition	Def. 1.1
<code>chainAdmissible_of_responseScaleControlled</code>	theorem	Thm. 1.4
<code>responseScaleControlledAt_identity</code>	theorem	Lem. 1.6
<code>responseScaleControlledAt_constant</code>	theorem	Lem. 1.6
<code>hasMulDerivativeAt_comp_of_responseScaleControlled</code>	theorem	Thm. 1.7

Every declaration of `PrimeTensor/Analysis/Growth.lean` appears above. The file contains no sorry, no axiom, and no unproved named proposition. Design note 1.2 and Remarks 1.3, 1.5, 1.8 and 1.9 are stated for the reader and are not declarations of the file; the numerical and logarithmic readings of the comparison are glosses and are not formalised.